Birthday Voter: How Birthdays Near Election Day Change Whether and How People Vote

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Abstract

A birthday in the days before Election Day lowers turnout, and a birthday on Election Day changes it little. We use voter files from Florida, Ohio, and Pennsylvania covering 76 statewide elections and 550,331,623 registrant-elections. Election dates move from year to year, so the same birth date falls on or near Election Day in some years and not others, and we compare people born on the same date across those years. A birthday the day before Election Day lowers turnout by 0.53 percentage points (SE 0.04), about 1.3% of the turnout rate, and the estimate is similar in each state. An Election Day birthday raises turnout by 0.13 points (95% CI 0.03 to 0.22), mostly in Ohio. The loss falls on voting in person. Where the files record how people voted, voting early or by mail rises around the birthday. We first saw the pattern in Florida's 2006–2016 elections, and it recurs in every sample not used to find it. Birthdays look less like an occasion for civic duty than like a claim on time that people plan around when the clash is exact but not when it is near.

1 Introduction

People are less likely to vote when their birthday falls in the days just before Election Day. Using voter files from three states, we find that a birthday the day before an election lowers turnout by about half a percentage point. A birthday on Election Day itself does not: people who would have voted at the polls that day vote early or by mail instead.

The pattern bears on why people vote. Few would vote if they voted only to change outcomes, because one vote almost never decides an election (Downs 1957; McLean and Hewitt 1994; but see Edlin, Gelman and Kaplan 2007). One explanation is that voting brings expressive benefits, among them the satisfaction of doing one's duty (Riker and Ordeshook 1968; Fiorina 1976; Kan and Yang 2001). Another is that voting costs little, so that small changes in its cost move turnout (Verba et al. 1993).

Birthdays separate these accounts. If people treat voting as a duty and like doing dutiful things on a day that feels significant, an Election Day birthday should raise turnout. If people take their birthday off work, voting should get cheaper, the logic behind proposals to make Election Day a holiday (Farber 2009), and turnout should again rise. If a birthday competes for time, with a dinner, a trip, or a day spent elsewhere, it should lower turnout or move votes to other days. The evidence fits the last account.

2 Data

Table 1 describes the data. For each state we have a voter file listing everyone registered when it was drawn, with their date of birth and registration date, and a history of their participation in past elections. The data are public records, distributed through the Harvard Dataverse; our replication materials contain counts by election and exact date of birth, with no names or identifiers.

Table 1: Data. Elections are statewide general elections and primaries; registrants are those with a date of birth in the state's largest extract; person-elections are registrant-elections in the analysis sample; turnout is the share of them who voted.

State	Elections	Registrants	Person-elections	Birthday on E-Day	Turnout	Method recorded
Florida	20 (2006–2020)	15,378,113	215,465,585	603,601	40.6%	polls, early, mail
Ohio	26 (2000–2016)	7,891,442	135,241,890	365,341	49.9%	–
Pennsylvania	30 (2009–2024)	8,823,823	199,624,148	538,260	34.8%	polls, mail

Florida. The February 2017 and June 2022 extracts of the Florida voter file record, for every election, whether a registrant voted at the polls on Election Day, voted early in person, voted by mail, or cast a ballot that was not counted. We take each election from the earliest extract filed after it, so a person on both lists contributes each election once. We use general elections, primaries, and presidential preference primaries from 2006 to 2020.

Ohio. The 2016 statewide voter file (with history through May 2017) has a column for each election since 2000; an entry means the person voted, and the file does not say how. We use every November general election, including the odd-year elections in which Ohio votes on statewide issues, and the spring primaries of even years.

Pennsylvania. The August 2024 Full Voter Export records each registrant's last 40 elections and how they voted: at the polls, absentee, by mail (from 2020), or provisionally. Pennsylvania has no early in-person voting. We use the 30 elections that every county's record covers, November 2009 to April 2024.

Two features of the files matter for the design. First, each file lists only people registered when it was drawn, so turnout in early elections is measured among those still on the rolls years later and runs above official figures (Appendix Table 4). This affects turnout levels, which the design absorbs, not the comparison of birthdays. Second, Ohio and Pennsylvania record a person's latest registration date, so we count someone as eligible for an election if that date clears the registration deadline or if they had voted in an earlier election.

3 Research Design

The unit is a registrant in an election, and the outcome is whether they voted. The quantity of interest is how turnout changes with the number of days between Election Day and the person's birthday that year, which we write d: $d = 0$ when Election Day is the birthday, $d = -1$ when the birthday was the day before.

People born at different times of year differ in many ways, so comparing people with $d = 0$ to everyone else would confound the birthday with the season of birth. Instead we compare people

born on the same calendar date in the same state across elections in which that date fell at different distances from Election Day. The comparison works because election dates move. General elections fall between November 2 and 8; Pennsylvania's primaries have fallen anywhere from April 22 to June 2; and Ohio's primaries move between March and May. A person born on November 4 has a birthday on Election Day in some years, a few days before or after it in others, and well away from it when they vote in a primary.

Formally, we regress whether registrant i voted in election e on indicators for each value of d from -30 to 30, leaving birthdays more than 30 days from Election Day as the comparison. The model includes fixed effects for each calendar birth date within a state and election type, which absorb everything constant about people born on a given date, and for each election within each birth year, which absorb differences in turnout by election and age. The coefficient on $d = -1$ is then the difference in turnout between people whose birthday was the day before an election and people born on the same date in elections where their birthday was far away, net of how turnout in that election differed for everyone born in their year.

The assumption is that, but for the birthday, the turnout of people born on a given date would rise and fall across elections like the turnout of others born the same year. Three things could violate it. Age is one, and the birth-year-by-election fixed effects absorb it. Registration is another. In Ohio and Pennsylvania, as in many states, driver's licences expire on birthdays, and renewing a licence can update a voter's registration; a change of address recorded just before an election could make it harder to vote. Registration dates in the Ohio file are indeed 4.8 times as common on birthdays as on days more than two months away. We test this explanation directly below. Finally, because d is the difference between a birth date and an election date, both of which have fixed effects, a linear trend in d is not identified (Borusyak, Jaravel and Spiess 2024); the comparison with birthdays far from Election Day pins down the level, and the sharp changes we report around the birthday cannot be produced by a linear trend.

To avoid comparing first-time voters with others, we drop any birth-year cohort whose youngest member turned 18 less than 60 days before an election, and cohorts aged 100 or more. These rules act on whole cohorts, so they never depend on where in the year someone was born.

Standard errors are clustered by calendar birth date (366 clusters), pooling across states. Treatment is assigned by birth date and election date, the same people and the same birth dates recur across elections, and people born on the same date may share calendar-specific shocks across states. We aggregate the data to counts by election and exact date of birth; a regression on cell shares weighted by cell size returns the same estimates and clustered standard errors as the person-level model. To summarize the daily estimates we report three quantities: the average effect over the seven days before the birthday, the effect of a birthday the day before Election Day, and the effect of a birthday on Election Day, with standard errors from the clustered covariance of the daily estimates.

We first saw the pattern in Florida's 2006–2016 elections. The other samples, Florida 2018–2020, Ohio, and Pennsylvania, were not used to find it, and we report them separately.

4 Results

Figure 1 shows the daily estimates for each state and for the three together. The pattern is the same in all three. Turnout begins to slip when the birthday falls two to three weeks before Election Day, falls fastest over the final week, and is lowest when the birthday is the day before. When the birthday is Election Day itself, turnout is back at or slightly above its usual level, and it stays there when the birthday falls after Election Day.

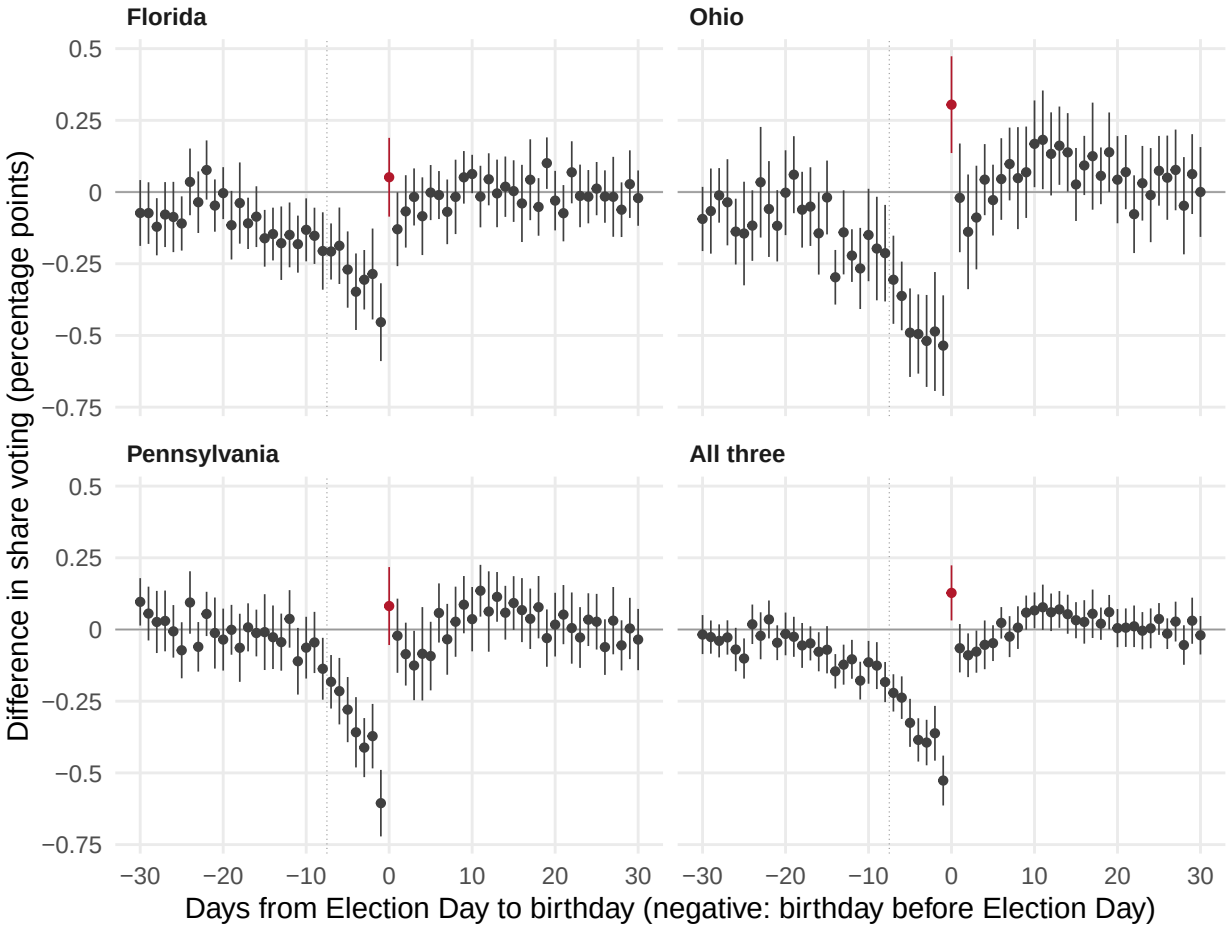


Figure 1: Turnout by days between birthday and Election Day. Estimates relative to birthdays more than 30 days from Election Day, with 95% confidence intervals; the red point is a birthday on Election Day, and the dotted line marks the week before it.

Table 2 summarizes the daily estimates. For all three states together, a birthday the day before Election Day lowers turnout by 0.53 points (SE 0.04), against a turnout of 40.4% among people with such birthdays, a fall of about 1.3%. The estimates for the three states are close: 0.45 points in Florida, 0.54 in Ohio, and 0.61 in Pennsylvania. Averaged over the week before the birthday, the fall is 0.35 points (SE 0.02); we chose the week after seeing Florida, and averaging over three days (0.43, SE 0.03) or two weeks (0.24, SE 0.02) tells the same story. An Election Day birthday, by contrast, raises turnout by 0.13 points (SE 0.05), an estimate driven by Ohio (0.30); in Florida

and Pennsylvania it is 0.05 and 0.08.

Table 2: Effects of a birthday near Election Day. Percentage points, with standard errors clustered by birth date in parentheses. Week before is the average over the seven days before the birthday.

	Outcome	Week before	Day before	Election Day
Florida	Turnout	−0.29 (0.03)	−0.45 (0.07)	0.05 (0.07)
	In person on Election Day	−0.26 (0.03)	−0.65 (0.07)	−0.39 (0.07)
	Early in person	−0.06 (0.02)	0.16 (0.04)	0.28 (0.05)
	By mail	0.03 (0.02)	0.03 (0.04)	0.16 (0.04)
Ohio	Turnout	−0.46 (0.06)	−0.54 (0.09)	0.30 (0.09)
Pennsylvania	Turnout	−0.35 (0.04)	−0.61 (0.06)	0.08 (0.07)
	In person on Election Day	−0.40 (0.04)	−0.71 (0.06)	−0.11 (0.09)
	By mail	0.05 (0.02)	0.10 (0.03)	0.19 (0.05)
All three	Turnout	−0.35 (0.02)	−0.53 (0.04)	0.13 (0.05)

How people vote changes more than whether they vote (Figure 2). In Florida and Pennsylvania the fall in turnout before a birthday is a fall in voting at the polls on Election Day: 0.65 and 0.71 points when the birthday is the day before. Voting at the polls is also lower for a few days after the birthday, when turnout as a whole is not, because voting early or by mail makes up for it. A birthday reduces voting in person on other days as well: Floridians whose birthday falls in the last week of early voting, a week or so before Election Day, vote early less often. When Election Day is the birthday, Floridians are 0.39 points less likely to vote at the polls but 0.28 points more likely to vote early and 0.16 points more likely to vote by mail, so turnout is unchanged. In Pennsylvania, which has no early in-person voting, voting by mail rises by 0.19 points.

Table 3 shows that the result does not depend on how standard errors are clustered; appears in general elections (0.81 points the day before) and primaries (0.45); and is as large in the samples that were not used to find it, 0.56 points (SE 0.05) the day before, as in the one that was (0.48). Nor is it the product of registration changes made at licence renewals. Among Ohio and Pennsylvania registrants whose latest registration came a year or more before the election, so that no registration change fell in the run-up to it, a birthday the day before lowers turnout by 0.58 points, against 0.58 among all registrants in the two states.

5 Interpretation

The evidence does not fit the idea that people treat an Election Day birthday as an occasion to do their duty. Turnout rises little, if at all, and in the two states that record how people vote, those with an Election Day birthday are less likely to vote at the polls, not more. Nor does it show that a birthday off work makes voting easier.

It fits better the idea that a birthday competes with voting in person, and that whether it costs a vote depends on whether people see the clash coming. Voting in person falls on the days around a birthday, whether that day is Election Day or a day of early voting. When Election Day is the birthday, the clash is obvious, and people who would have voted at the polls vote early or by mail

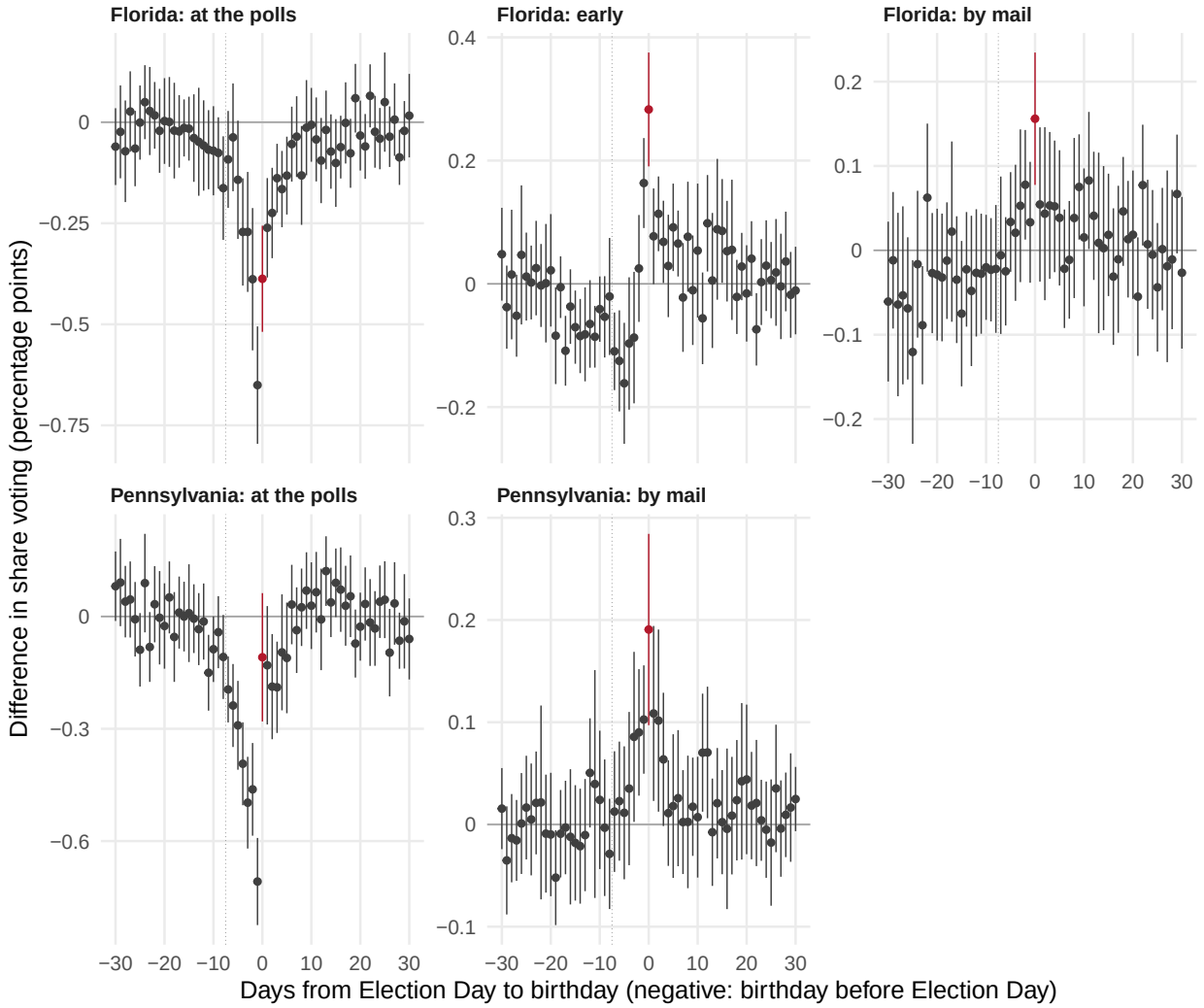


Figure 2: Voting by method and days between birthday and Election Day. Florida records voting at the polls, early, and by mail; Pennsylvania records voting at the polls and by mail or absentee ballot (shown together) and has no early in-person voting. Ohio does not record how people voted. Estimates and intervals as in Figure 1.

Table 3: Robustness of the turnout estimates for the three states together. Percentage points, standard errors in parentheses, clustered by birth date unless noted. [a]The two-way clustered covariance matrix was not positive definite and was adjusted by setting negative eigenvalues to zero.

	Week before	Day before	Election Day
Main specification	−0.35 (0.02)	−0.53 (0.04)	0.13 (0.05)
Clustered by birth date within state	−0.35 (0.02)	−0.53 (0.04)	0.13 (0.05)
Clustered by birth date and by election[a]	−0.35 (0.03)	−0.53 (0.05)	0.13 (0.05)
Florida 2006–2016 only	−0.33 (0.04)	−0.48 (0.08)	0.06 (0.09)
All other elections	−0.38 (0.03)	−0.56 (0.05)	0.17 (0.05)
General elections only	−0.64 (0.06)	−0.81 (0.08)	−0.04 (0.07)
Primaries only	−0.30 (0.03)	−0.45 (0.05)	0.10 (0.07)
Ohio and Pennsylvania	−0.39 (0.03)	−0.58 (0.05)	0.17 (0.06)
registered a year or more before	−0.39 (0.03)	−0.58 (0.05)	0.13 (0.06)

instead, so turnout holds up. When the birthday falls a few days before Election Day, the clash is less obvious. The celebration, the trip, or the day off comes first, Election Day arrives while it is still going on or just after, and some of those who would have voted in person do not vote at all.

The data cannot say what people do on and around their birthdays, so this account is an interpretation rather than a finding. Three features of the evidence support it: the loss is in voting in person and not by mail; it is concentrated in the days just before Election Day rather than spread over the month; and voting early or by mail rises when the birthday is Election Day. Its leading mechanical alternative, registration changes made at driver's licence renewals, does not account for the pattern. The asymmetry is also hard to square with explanations that do not involve the birthday as an event, such as seasonality, which the design absorbs.

The effects are small. A fall of half a percentage point among the roughly one in 365 registrants whose birthday is the day before an election changes no election. But they show that turnout responds to modest, predictable demands on a person's time in the days around an election, and that the option to vote on another day absorbs some of that cost when people anticipate it.

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A Turnout by Election

Table 4: Registrants and turnout by election. Turnout is the share of registrants in the analysis sample who voted; Official is the state's reported turnout where we could verify it.

State	Election	Registrants	Turnout	Official	Birthday on E-Day
Florida	Sep 5, 2006 PRI	7,275,221	21.0%	20%	20,771
	Nov 7, 2006 GEN	7,346,234	50.7%	47%	20,500
	Jan 29, 2008 PPP	7,804,524	43.8%	42%	20,494
	Aug 26, 2008 PRI	8,213,432	18.2%	18%	23,695
	Nov 4, 2008 GEN	8,575,718	80.1%	75%	24,136
	Aug 24, 2010 PRI	9,216,016	22.6%	22%	26,430
	Nov 2, 2010 GEN	9,299,145	50.9%	49%	25,845
	Jan 31, 2012 PPP	9,790,560	16.5%	41%	25,652
	Aug 14, 2012 PRI	10,189,899	20.7%	21%	29,246
	Nov 6, 2012 GEN	10,507,462	73.7%	72%	28,934
	Aug 26, 2014 PRI	11,503,073	17.0%	18%	33,069
	Nov 4, 2014 GEN	11,616,728	49.5%	51%	32,529
	Mar 15, 2016 PPP	12,519,303	32.5%	46%	34,295
	Aug 30, 2016 PRI	12,894,121	22.5%	24%	36,991
	Nov 8, 2016 GEN	13,174,364	70.8%	75%	36,841
	Aug 28, 2018 PRI	12,082,114	26.9%	27%	35,121
	Nov 6, 2018 GEN	12,267,419	62.3%	63%	34,073
	Mar 17, 2020 PPP	13,381,904	21.3%	30%	35,946
	Aug 18, 2020 PRI	13,677,239	27.1%	28%	39,482
	Nov 3, 2020 GEN	14,131,109	75.5%	77%	39,551
Ohio	Mar 7, 2000 PRI	2,437,416	47.6%	–	6,585
	Nov 7, 2000 GEN	2,841,507	86.3%	–	7,731
	Nov 6, 2001 GEN	3,464,860	47.7%	–	9,393
	May 7, 2002 PRI	3,550,654	24.6%	–	9,299
	Nov 5, 2002 GEN	3,608,309	60.7%	–	9,814
	Nov 4, 2003 GEN	3,757,964	43.5%	–	10,328
	Mar 2, 2004 PRI	3,841,226	41.4%	–	10,365
	Nov 2, 2004 GEN	4,077,784	90.9%	–	11,112
	Nov 8, 2005 GEN	4,693,490	50.4%	–	12,705
	May 2, 2006 PRI	4,763,348	30.0%	–	12,520
	Nov 7, 2006 GEN	4,820,526	67.4%	–	13,064
	Nov 6, 2007 GEN	4,978,821	34.9%	–	13,471
	Mar 4, 2008 PRI	5,057,829	57.4%	–	13,818
	Nov 4, 2008 GEN	5,284,519	86.2%	69.97%	14,450
	Nov 3, 2009 GEN	5,674,587	50.0%	–	15,400
	May 4, 2010 PRI	5,766,412	27.2%	–	15,038
	Nov 2, 2010 GEN	5,829,994	59.6%	–	15,768
	Nov 8, 2011 GEN	6,017,965	54.5%	–	16,281
	Mar 6, 2012 PRI	6,124,785	29.2%	–	16,495
	Nov 6, 2012 GEN	6,312,807	79.8%	70.53%	17,115

State	Election	Registrants	Turnout	Official	Birthday on E-Day
	Nov 5, 2013 GEN	6,684,151	29.2%	–	18,103
	May 6, 2014 PRI	6,787,236	18.2%	–	17,807
	Nov 4, 2014 GEN	6,873,345	43.7%	–	18,578
	Nov 3, 2015 GEN	7,125,003	44.2%	–	19,261
	Mar 15, 2016 PRI	7,269,254	44.3%	–	20,137
	Nov 8, 2016 GEN	7,598,098	71.8%	71.33%	20,703
Pennsylvania	Nov 3, 2009 GEN	5,089,456	24.6%	–	13,712
	May 18, 2010 PRI	5,140,352	24.5%	–	13,955
	Nov 2, 2010 GEN	5,183,081	50.9%	–	13,972
	May 17, 2011 PRI	5,259,914	20.6%	–	14,033
	Nov 8, 2011 GEN	5,305,951	27.0%	–	14,452
	Apr 24, 2012 PRI	5,378,732	20.2%	–	14,182
	Nov 6, 2012 GEN	5,503,735	70.3%	67.63%	15,013
	May 21, 2013 PRI	5,684,730	15.3%	–	15,096
	Nov 5, 2013 GEN	5,729,058	22.9%	–	15,780
	May 20, 2014 PRI	5,791,363	17.1%	–	15,431
	Nov 4, 2014 GEN	5,843,440	45.3%	–	15,720
	May 19, 2015 PRI	5,929,854	20.0%	–	15,812
	Nov 3, 2015 GEN	6,000,477	29.4%	–	16,088
	Apr 26, 2016 PRI	6,157,174	42.9%	–	16,200
	Nov 8, 2016 GEN	6,437,720	73.8%	70.11%	17,524
	May 16, 2017 PRI	6,723,220	17.7%	–	18,151
	Nov 7, 2017 GEN	6,772,403	27.3%	–	18,422
	May 15, 2018 PRI	6,858,484	19.7%	–	18,563
	Nov 6, 2018 GEN	6,959,206	60.6%	–	18,979
	May 21, 2019 PRI	7,157,787	20.6%	–	19,186
	Nov 5, 2019 GEN	7,244,125	33.6%	–	19,913
	Jun 2, 2020 PRI	7,454,732	33.5%	–	20,372
	Nov 3, 2020 GEN	7,836,509	76.6%	76.07%	21,111
	May 18, 2021 PRI	8,017,039	27.2%	–	21,795
	Nov 2, 2021 GEN	8,074,129	32.7%	–	21,652
	May 17, 2022 PRI	8,167,450	31.1%	–	22,026
	Nov 8, 2022 GEN	8,306,757	61.4%	–	22,516
	May 16, 2023 PRI	8,438,870	24.2%	–	22,795
	Nov 7, 2023 GEN	8,519,314	36.5%	–	23,052
	Apr 23, 2024 PRI	8,659,086	24.1%	–	22,757